

# YOLO Model Testing Report

|                      |                                           |
|----------------------|-------------------------------------------|
| Model:               | yolov8n                                   |
| Model Size:          | 6.2 MB                                    |
| Parameters:          | 3.2 M                                     |
| FLOPs (640x640):     | 8.86 GFLOPs                               |
| Run Name:            | yolov8n_testing_20251123_223435           |
| W&B Run Name:        | yolov8n_bdd100k_yolo_test_20251123_223435 |
| Timestamp:           | 2025-11-23 22:38:13                       |
| Dataset:             | bdd100k_yolo - test split                 |
| Images Processed:    | 20000                                     |
| Total Execution Time | 197.31s                                   |
| IoU Threshold:       | 0.5                                       |

## Inference Performance

| Metric                  | Value             |
|-------------------------|-------------------|
| Average Inference Time  | 4.85 ms per image |
| FPS (Frames Per Second) | 206.22            |

Note: Inference time includes preprocessing, model inference, and postprocessing.

## Overall Accuracy Metrics

| Metric    | Value  |
|-----------|--------|
| Precision | 0.2308 |
| Recall    | 0.6309 |
| F1-Score  | 0.3380 |
| mAP@0.5   | 0.0788 |

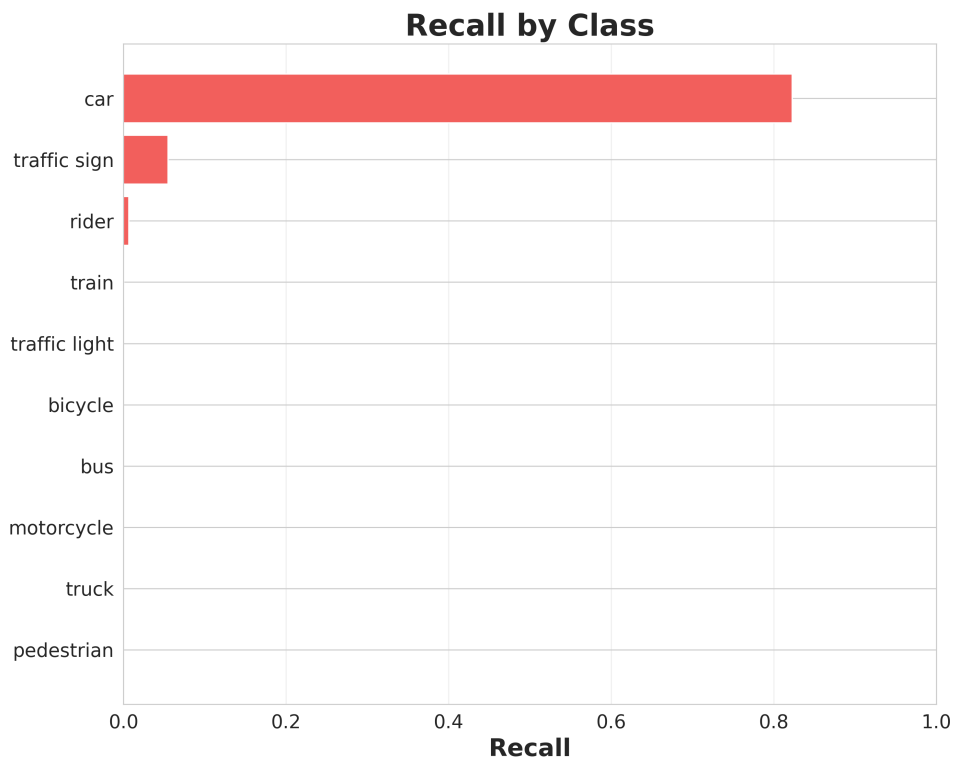
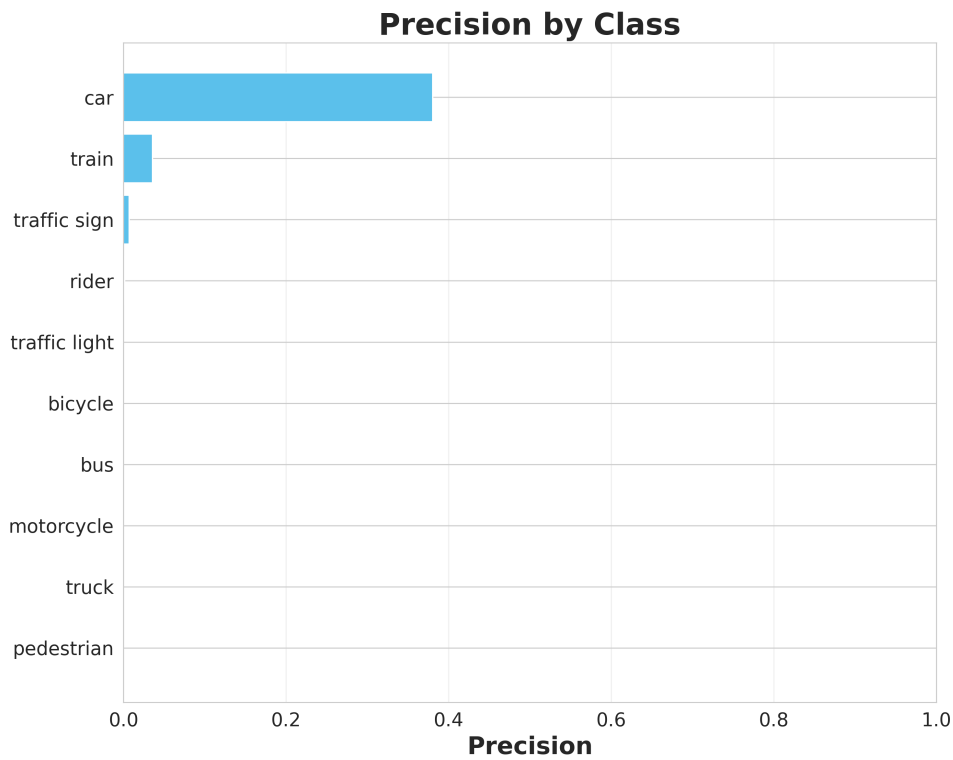
|                 |        |
|-----------------|--------|
| mAP@0.5:0.95    | 0.0436 |
| True Positives  | 78520  |
| False Positives | 261621 |
| False Negatives | 45932  |

# Confusion Matrix

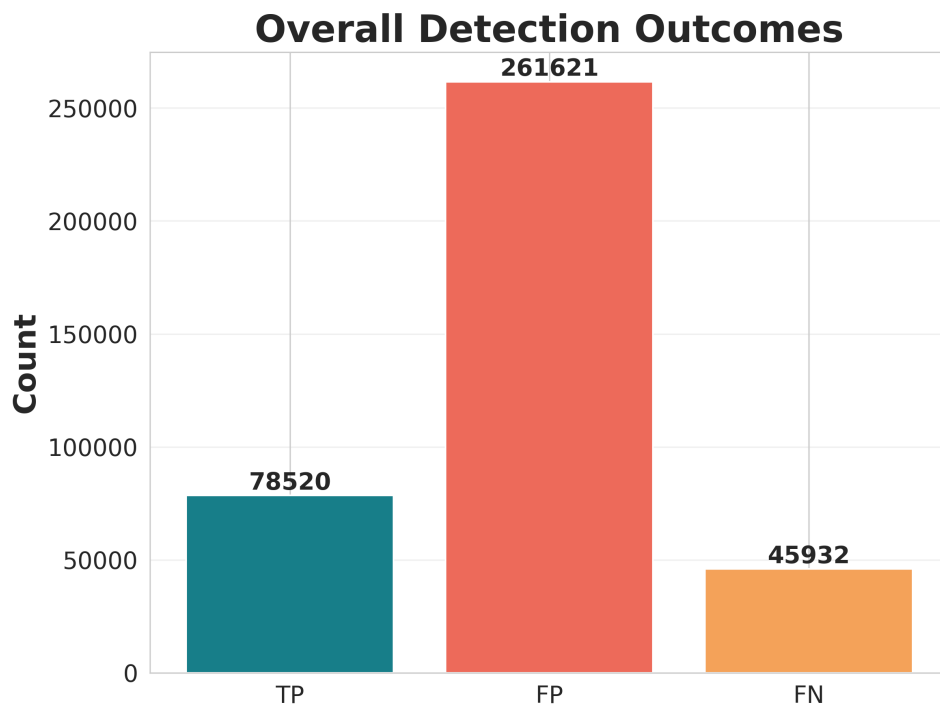
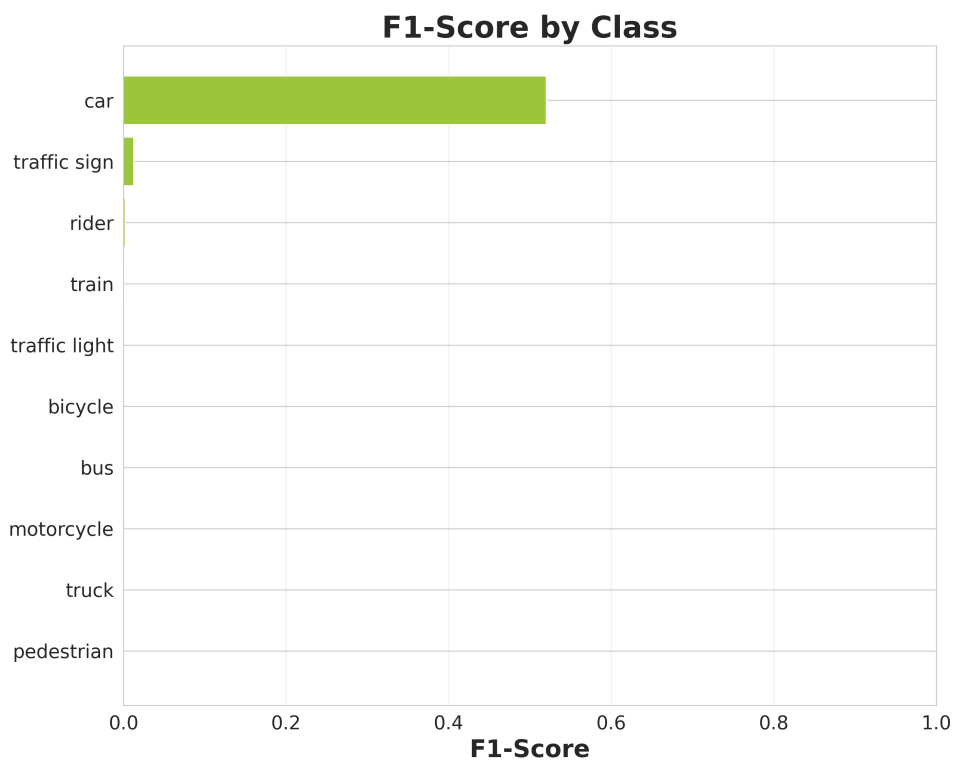
| Confusion Matrix (yolov8n validation) |               |            |       |      |       |     |       |            |         |               |              |
|---------------------------------------|---------------|------------|-------|------|-------|-----|-------|------------|---------|---------------|--------------|
| True Class                            | pedestrian    | 449        | 82    | 2    |       |     |       |            | 4       | 36            |              |
|                                       | rider         | 2          | 1     |      |       |     |       |            |         |               |              |
|                                       | car           | 18         | 78024 | 1188 | 225   |     |       |            | 19      | 70            |              |
|                                       | truck         | 11         | 14    |      |       |     |       |            |         |               |              |
|                                       | bus           |            | 3     | 1    |       |     |       |            |         |               |              |
|                                       | train         |            | 386   | 359  | 822   | 1   |       |            |         | 5             |              |
|                                       | motorcycle    |            | 16    | 29   | 13    | 4   |       |            |         |               |              |
|                                       | bicycle       | 2          | 2360  | 2213 | 359   |     |       |            | 1       | 10            |              |
|                                       | traffic light |            | 6     | 1    |       |     |       |            |         |               |              |
|                                       | traffic sign  |            | 2     | 1    |       | 2   |       |            | 4765    | 493           |              |
|                                       |               | pedestrian | rider | car  | truck | bus | train | motorcycle | bicycle | traffic light | traffic sign |
| Predicted Class                       |               |            |       |      |       |     |       |            |         |               |              |

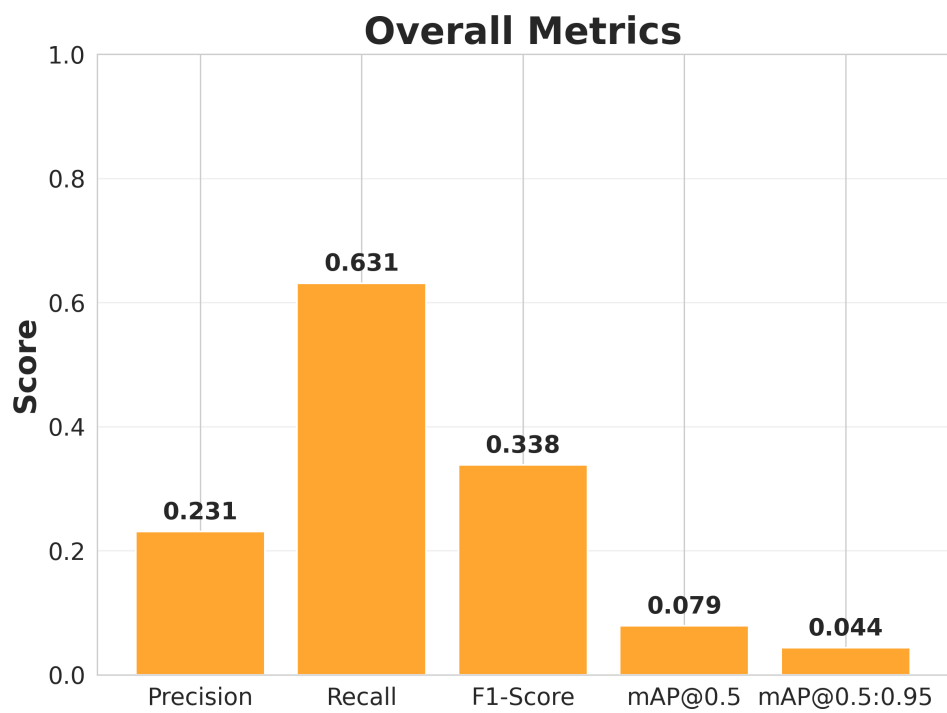
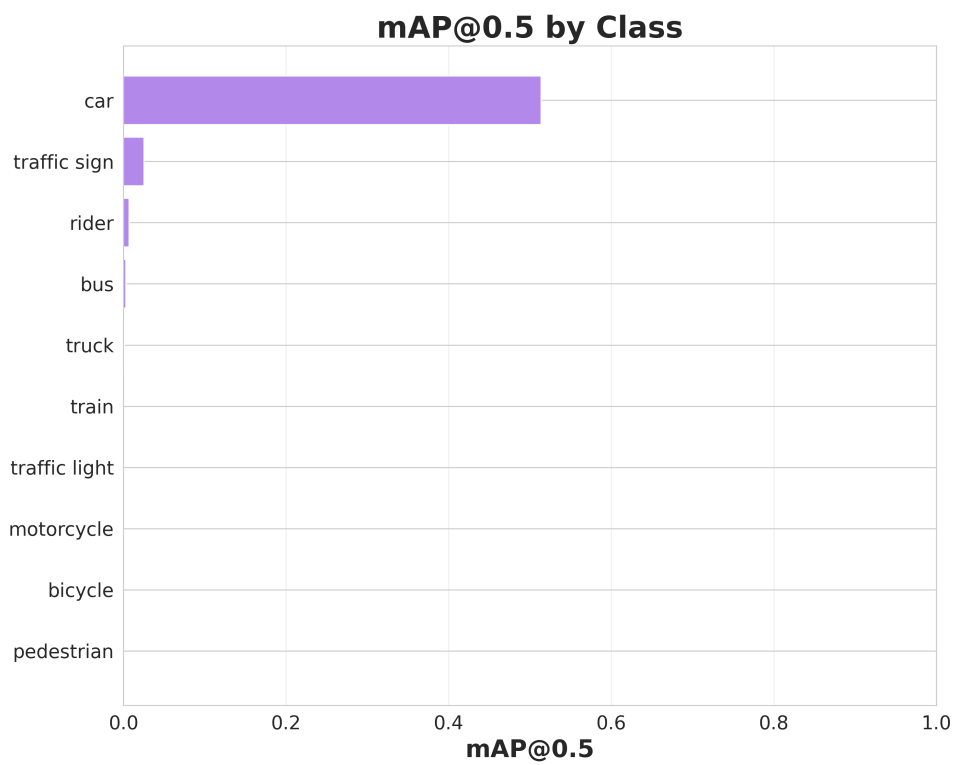
Correct Predictions (Diagonal Sum): 78520  
Total Matched Predictions: 373315

# Performance Visualizations









# Per-Class Performance

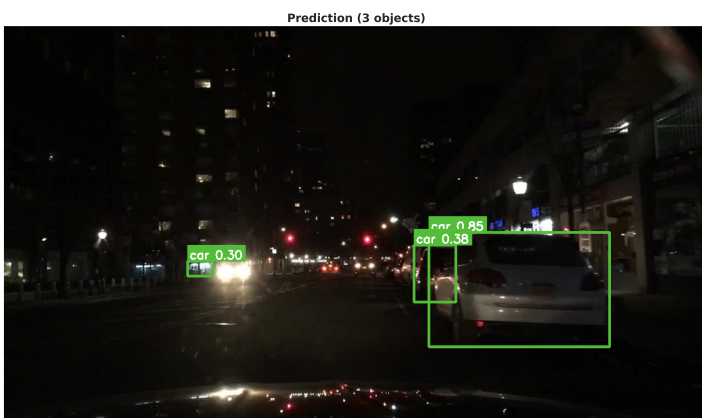
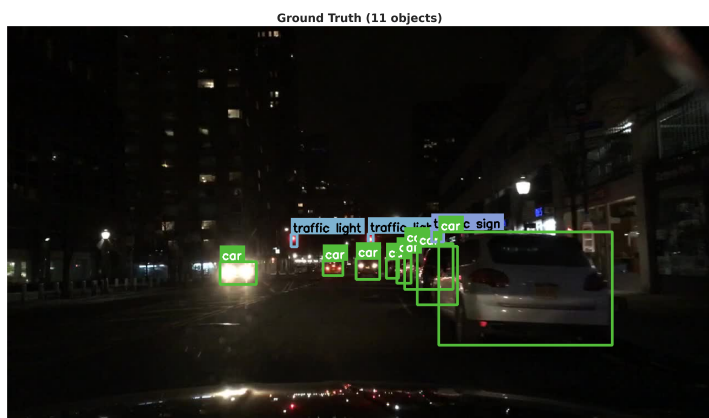
| Class         | TP    | FP     | FN    | Precision | Recall | F1-Score | mAP@0.5 |
|---------------|-------|--------|-------|-----------|--------|----------|---------|
| pedestrian    | 0     | 0      | 9887  | 0.0000    | 0.0000 | 0.0000   | 0.0000  |
| rider         | 2     | 1291   | 298   | 0.0015    | 0.0067 | 0.0025   | 0.0072  |
| car           | 78024 | 127069 | 16799 | 0.3804    | 0.8228 | 0.5203   | 0.5142  |
| truck         | 0     | 8701   | 229   | 0.0000    | 0.0000 | 0.0000   | 0.0013  |
| bus           | 0     | 3217   | 35    | 0.0000    | 0.0000 | 0.0000   | 0.0030  |
| train         | 1     | 27     | 2410  | 0.0357    | 0.0004 | 0.0008   | 0.0002  |
| motorcycle    | 0     | 0      | 238   | 0.0000    | 0.0000 | 0.0000   | 0.0000  |
| bicycle       | 0     | 0      | 7552  | 0.0000    | 0.0000 | 0.0000   | 0.0000  |
| traffic light | 0     | 52817  | 27    | 0.0000    | 0.0000 | 0.0000   | 0.0001  |
| traffic sign  | 493   | 68499  | 8457  | 0.0071    | 0.0551 | 0.0127   | 0.0255  |

# Sample Predictions: Ground Truth vs Model

## Sample #1: f4132bdf-956437ae.jpg

Ground Truth Objects: 11 | Predicted Objects: 3

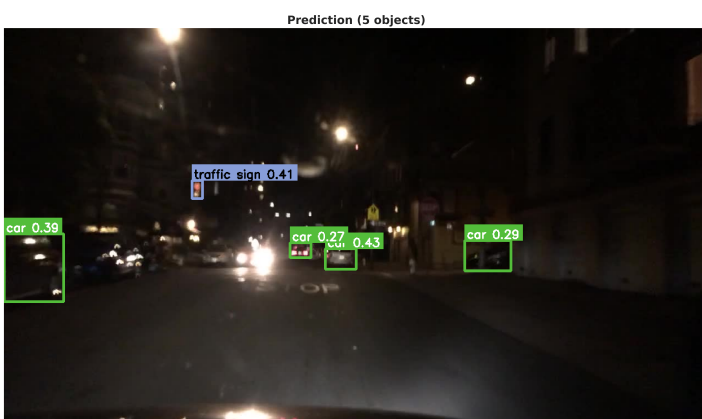
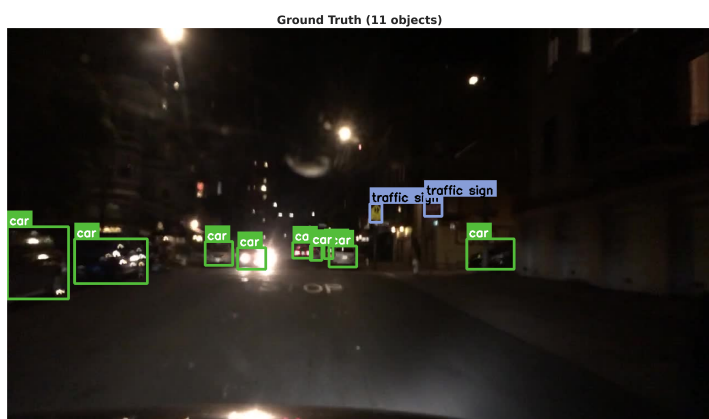
Weather: clear | Scene: city street | Time of Day: night



## Sample #2: e44b4ba9-766c80e0.jpg

Ground Truth Objects: 11 | Predicted Objects: 5

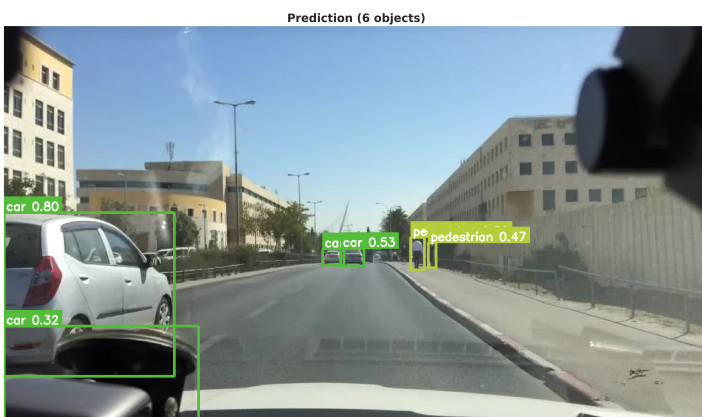
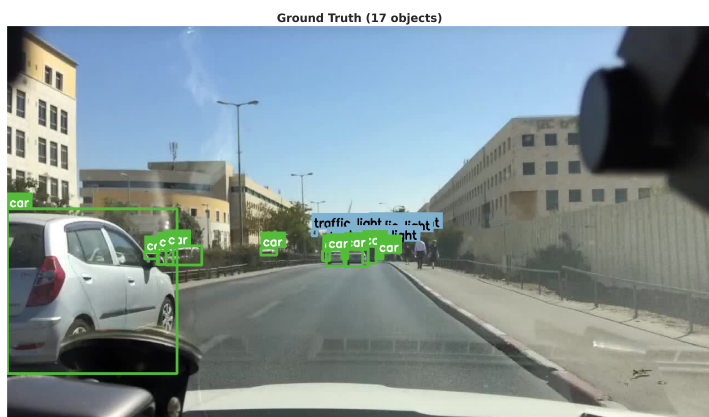
Weather: clear | Scene: city street | Time of Day: night



## Sample #3: ced5759e-f51a32d9.jpg

Ground Truth Objects: 17 | Predicted Objects: 6

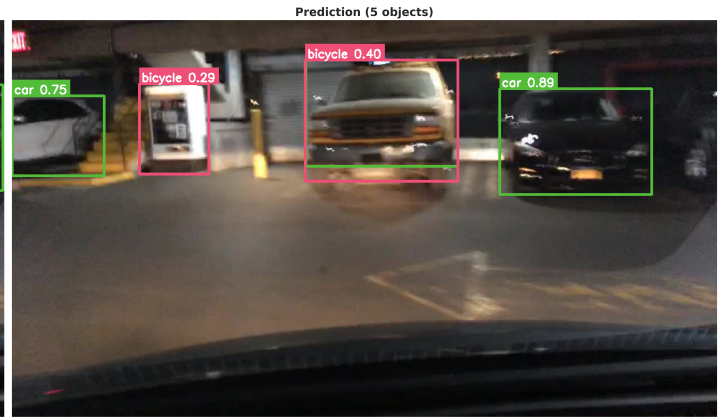
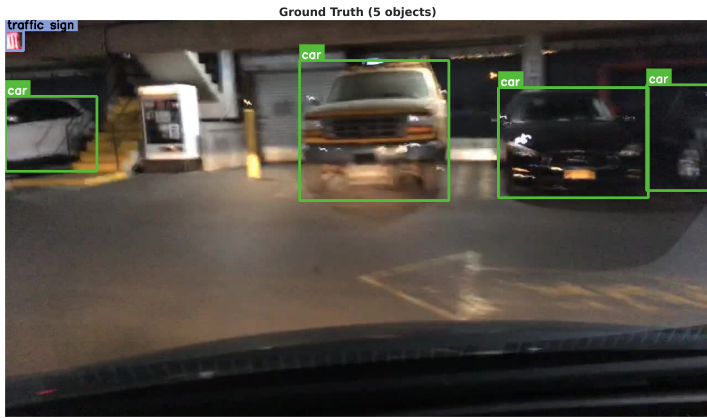
Weather: clear | Scene: city street | Time of Day: daytime



## Sample #4: ced33a3e-1d3a752e.jpg

Ground Truth Objects: 5 | Predicted Objects: 5

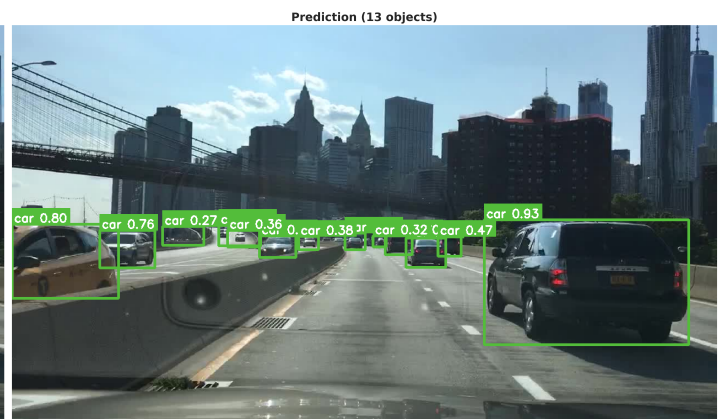
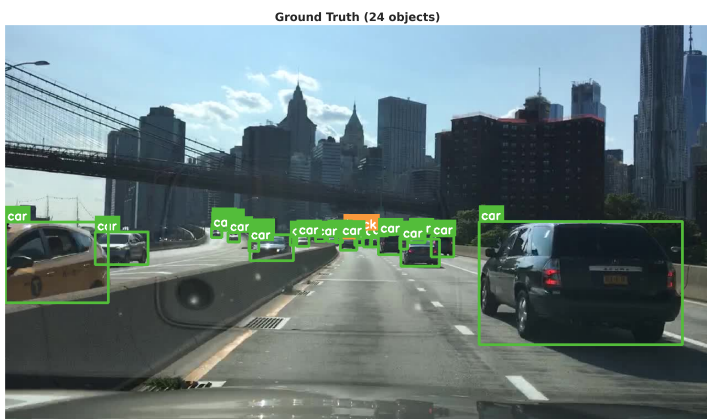
Weather: undefined | Scene: parking lot | Time of Day: night



### Sample #5: f4683191-14c2cfe3.jpg

Ground Truth Objects: 24 | Predicted Objects: 13

Weather: partly cloudy | Scene: highway | Time of Day: daytime



### Sample #6: d97776a2-e591c99a.jpg

Ground Truth Objects: 16 | Predicted Objects: 7

Weather: overcast | Scene: city street | Time of Day: dawn/dusk



Generated by YOLO Quick Test Script